Q07: Discussion Questions

DLS 2026 — Kukla

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1. Scaling laws let you predict a model's loss before training it. What are the business and safety implications of being able to forecast capability from a budget?
2. Chinchilla says scale N and D together, but high-quality data is finite. Rank the proposed escapes (synthetic data, multi-epoch, data curation, new modalities) by how much you trust them.
3. A smooth loss curve can coexist with abrupt downstream capability jumps. Is "emergence" then a property of the model or of our metrics? Defend a view.
4. Classical learning theory fails to explain why over-parameterized models scale gracefully. Does this mean the theory is wrong, or incomplete, or measuring the wrong thing?
5. Recent gains come increasingly from post-training and inference-time compute. Does this mean the pre-training scaling era is ending, or just that we found additional axes? Argue.